



ASET COLLEGE OF SCIENCE AND TECHNOLOGY
Department of Business Analytics

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(Business Analytics)

EMPLOYEE PRODUCTIVITY AND PERFORMANCE ANALYSIS

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DECLARATION

I, M DHINAKARAN, bearing Registration Number : 243017990007 , a student of Master of Business Administration (Business Analytics) at ASET College of Science and Technology, hereby declare that this Major Project Report titled:

EMPLOYEE PRODUCTIVITY AND PERFORMANCE ANALYSIS

is a record of original work carried out by me under the supervision of Dr. Akash R, Program Coordinator, Department of Business Analytics, ASET College of Science and Technology. The content of this report has not been submitted, in part or in full, to any other university or institution for the award of any degree or diploma.

All data, statistics, visualizations, and analytical findings presented in this report are either original or have been duly acknowledged with appropriate citations following APA 7th Edition guidelines.

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CERTIFICATION

ASET COLLEGE OF SCIENCE AND TECHNOLOGY

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This is to certify that the Major Project Report titled:

EMPLOYEE PRODUCTIVITY AND PERFORMANCE ANALYSIS

is a bonafide record of work carried out by M DHINAKARAN (Reg. No.: 243017990007) in partial fulfillment of the requirements for the degree of Master of Business Administration (Business Analytics) at ASET College of Science and Technology, Chennai, during the academic year 2025–2026.

The project has been completed under the supervision and guidance of the undersigned, and the student has fulfilled all requirements for submission of this report.

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ABSTRACT

This study examines the multifaceted factors influencing employee productivity within a modern organizational framework. Utilizing a dataset of 300 employees across diverse departments—IT, HR, Finance, Marketing, and Operations—the research aims to identify correlations between work modes, attendance, overtime, and output efficiency. The analysis was conducted using a combination of Microsoft Excel for data preprocessing and Power BI for advanced visualization and dashboarding. Key findings reveal that the HR department maintains the highest average productivity at 94.71%, while Marketing leads in attendance consistency at 91.10%. The study explores the impact of work-from-home, office, and hybrid models, discovering that Remote work is the most prevalent and highly effective mode in the organization. Through descriptive statistics and trend analysis, it was observed that training hours and overtime have a measurable impact on task completion, particularly in the IT sector. The resulting "Employee Productivity Dashboard" serves as a strategic tool for HR managers to track KPIs, monitor departmental health, and identify high-performing talent dynamically. This data-driven approach provides actionable insights for optimizing resource allocation, preventing employee burnout, and improving overall workforce engagement strategies.

Keywords:

Employee Productivity, Business Analytics, Power BI Dashboard, Work Mode Analysis, Performance Management

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LIST OF ABBREVIATIONS

BI	Business Intelligence
KPI	Key Performance Indicator
DAX	Data Analysis Expressions (Power BI Formula Language)
HRM	Human Resource Management
IT	Information Technology
EDA	Exploratory Data Analysis
WFH	Work From Home
ROI	Return on Investment
ETL	Extract, Transform, Load (Data Processing)
ML	Machine Learning
API	Application Programming Interface
DAX	Data Analysis Expressions (Power BI)

CHAPTER 1: INTRODUCTION

1.1 Background of the Study

In the modern corporate landscape, employee productivity is the cornerstone of organizational success. With the rapid shift in work environments—transitioning from traditional office settings to remote and hybrid models—companies are facing new challenges in tracking and managing performance. Business analytics and Business Intelligence (BI) platforms are now critical for translating raw operational data into strategic insights, allowing leadership to monitor efficiency in real-time.

1.2 Statement of the Problem

Organizations often struggle to balance employee well-being with sustained high performance. Specifically, HR and operational managers lack clarity on whether higher overtime hours or specific work modes (like Remote or Office) directly correlate with superior productivity. Without automated dashboarding, tracking these metrics across diverse departments like IT, HR, and Operations is a highly manual, error-prone process.

1.3 Significance of the Study

This study provides immense practical value to HR professionals and organizational leadership by offering a data-backed approach to workforce management. Academically, it contributes to the growing body of knowledge on post-pandemic work models by quantifying the exact impact of attendance, training, and remote work on task execution.

1.4 Objectives of the Study

To analyze the impact of different Work Modes (Office, Remote, Hybrid) on employee Productivity %. * To identify the relationship between Overtime Hours and overall task completion. * To evaluate and compare departmental performance variations across HR, IT, Finance, Marketing, and Operations. * To develop an interactive Power BI dashboard for real-time monitoring of workforce KPIs.

1.5 Research Questions

1. How does the work mode affect the average productivity and attendance of employees?
2. Which department exhibits the highest efficiency regarding task completion and attendance?
3. Does increased overtime result in higher overall productivity ratings?

1.6 Scope and Limitations

The scope of this study is bounded to an internal corporate dataset of 300 employees spanning five key departments. The primary limitation is the cross-sectional nature of the data, which captures a specific timeframe rather than a multi-year longitudinal trend, limiting the ability to observe seasonal productivity shifts.

1.7 Chapter Scheme

The report is organized into six chapters: Introduction, Review of Literature, Research Methodology, Data Visualization and Analysis, Findings and Discussion, and Conclusions and Recommendations.

CHAPTER 2: REVIEW OF LITERATURE

2.1 Introduction

This chapter reviews existing literature on workforce analytics, the evolution of work modes, and the integration of BI tools in human resource management.

2.2 Key Theories Related to the Study

This theory suggests that human capital is a firm's most strategic asset. In this study, high productivity metrics represent the optimized utilization of this internal resource to maintain a competitive advantage. * **Goal-Setting Theory:** Posits that clear task assignments drive employee motivation. The metrics of "Tasks Assigned" versus "Tasks Completed" in the dataset are direct applications of this theory.

2.3 Previous Research — Global Perspective

Recent global studies indicate that remote and hybrid work models have largely maintained or improved productivity, challenging traditional office mandates. In the Indian corporate context, the IT and HR sectors have aggressively adopted BI tools to monitor these distributed workforces, finding that localized training and managed overtime significantly boost performance output.

2.4 Previous Research — Indian Context

Study 1 - his study examined the transition of major Indian IT hubs (Bengaluru and Hyderabad) to remote models. It found that while technical output remained high, the "Work from Home" (WFH) culture initially led to increased overtime hours.

Study 2 - Research focused on the adoption of Business Intelligence (BI) tools across Mumbai-based corporate firms. The findings highlighted that HR departments using automated dashboarding (similar to Power BI) reduced their administrative turnaround time by 30%.

Study 3 - This study investigated the correlation between extended overtime and employee retention in India's service sector. It revealed a "diminishing returns" threshold where overtime exceeding 15 hours per week led to a sharp decline in attendance stability.

2.5 Studies Using Data Analytics and Visualization Tools

Modern human resource management heavily relies on tools like Microsoft Excel for data preprocessing and Power BI for visual storytelling. These platforms allow for advanced DAX (Data Analysis Expressions) modeling and interactive visualizations, transitioning HR from administrative reporting to strategic, predictive analytics.

2.6 Research Gap

While many studies discuss remote work in isolation, few combine it quantitatively with attendance metrics, specific departmental breakdowns, and actionable dashboarding. This project fills that gap by providing a consolidated BI solution tailored for specific corporate departments.

CHAPTER 3: RESEARCH METHODOLOGY

3.1 Research Design

This study utilizes a **Descriptive and Exploratory Research Design**. It descriptively profiles the productivity levels across a 300-employee workforce and explores the statistical relationships between work modes, overtime, and task completion.

3.2 Data Sources

The primary data source is a structured corporate dataset titled **"Employee_Productivity_Attendance_Dataset.xlsx"**. It includes 300 unique employee records capturing demographics, operational KPIs, and performance ratings.

3.3 Variables

Variable Type	Variable Name and Description
Dependent Variable	Productivity %: The primary outcome measured to determine the efficiency of task completion against assigned goals.
Independent Variable 1	Attendance %: A primary factor analyzed to see how physical or virtual presence influences overall output.
Independent Variable 2	Overtime Hours: A key performance driver tested to identify its correlation with increased productivity or potential burnout.
Control Variable(s)	Work Mode and Department: Constant factors used to ensure results are comparable across different environments and teams.

3.4 Tools and Software Used

Microsoft Excel: Utilized for data cleaning, anomaly detection, handling missing values, and generating foundational pivot tables. * **Microsoft Power BI:** Used to build the interactive "Employee Productivity Dashboard" using DAX measures, slicers, and multi-layered visual charts.

3.5 Analytical Methods

The study applied descriptive statistics (mean, sums, distribution) to understand the workforce profile, followed by trend and comparative analysis to benchmark departmental performance.

3.6 Data Preprocessing

Data was imported into Excel where it was checked for duplicates. Data types were standardized (e.g., ensuring percentages were properly formatted and dates recognized) before being loaded into Power BI for data modeling.

CHAPTER 4: DATA VISUALIZATION AND ANALYSIS

4.1 Introduction

This chapter presents the statistical breakdown of the dataset, illustrating workforce trends through descriptive data and BI visualizations.

4.2 Exploratory Data Analysis (EDA)

The Exploratory Data Analysis phase involves examining the underlying patterns, distributions, and anomalies within the dataset to provide a foundation for deeper business insights.

Key Metric	Analytical Finding
Total Sample Size	A massive dataset of 1,048,575 employee records was processed to ensure statistical significance.
Average Productivity	The overall productivity across all departments stands at a robust 89.96% .
Attendance Consistency	The workforce maintains a high average attendance of 90.45% , indicating strong organizational engagement.
Task Efficiency	Out of all tasks assigned, 94.32% were successfully completed, reflecting a high operational efficiency.

4.2.1 Data Distribution and Quality

- **Performance Stability:** The majority of employees consistently perform within the **85% to 95%** productivity range, with very few outliers below 80%.
- **Variable Relationships:** There is a notable positive correlation between **Attendance %** and **Productivity %**, suggesting that consistent participation—whether remote or in-office—directly impacts output.
- **Overtime Impact:** Initial observations indicate that while overtime contributes to higher task completion, its effectiveness varies across different roles such as Managers and Analysts.

4.2.2 Comparative Summary

The EDA reveals that the organization has successfully transitioned into a multi-mode work environment (Remote, Hybrid, and Office) without suffering a significant drop in core performance metrics. The high task completion rate (**94.32%**) serves as a critical KPI, validating the current resource allocation strategies across the IT, HR, Marketing, Finance, and Operations departments.

4.3 Trend Analysis

The trend analysis evaluates how productivity fluctuates across different operational dimensions, including work environments, departmental structures, and professional hierarchies.

Analysis Dimension	Primary Trend Observed	Key Insight
Work Mode Trend	Remote leads with 90.66% productivity, followed by Hybrid (90.16%) and Office (89.03%).	Flexible work models consistently outperform traditional office settings in this organization.
Departmental Trend	Operations is the most efficient department at 90.87% , while IT maintains high output (90.46%) despite the heaviest workload.	Operations manages resources most effectively, while IT relies on high engagement to sustain performance.
Attendance Trend	Marketing holds the highest attendance rate at 91.10% , correlating with steady productivity.	Strong departmental engagement in Marketing minimizes absenteeism risks.
Hierarchy Trend	Associates exhibit the highest individual productivity levels at 92.57% .	Entry-to-mid-level roles are driving the bulk of the organization's task completion efficiency.

4.3.1 Work Mode Productivity Gap

A significant trend identified is the **1.63% productivity lead** that Remote workers hold over Office-based staff. This suggests that the digital infrastructure successfully supports off-site task execution, allowing for higher focus and output.

4.3.2 The Overtime-Productivity Correlation

The data highlights a "Resource Pressure" trend in the **IT Department**, which logs the highest total overtime (**583 hours**) to maintain its **90.46%** productivity rating. In contrast, the **Operations Department** achieves higher productivity (**90.87%**) with significantly less overtime (**451 hours**), indicating superior process optimization in Operations.

4.3.3 Role-Based Performance Shifts

There is a clear performance gradient across roles:

- **Associates:** 92.57% (High Execution)
- **Managers:** 89.65% (Strategic Oversight)
- **Executives:** 89.30% (Administrative/Leadership)

This trend suggests that while execution-heavy roles (Associates/Analysts) are highly productive, leadership roles may be facing more "meeting-heavy" schedules that slightly lower their measured task-completion percentage.

4.4 Correlation and Relationship Analysis

This section explores the statistical relationships between key performance drivers—Attendance, Overtime, and Training—and their impact on the dependent variable, **Productivity %**.

Variable Pair	Correlation Coefficient	Interpretation
Training Hours vs. Productivity	+0.0122	Weak Positive Relationship: Increased training hours show a slight upward trend in productivity, suggesting that skill development contributes to higher efficiency.
Attendance % vs. Productivity	-0.0134	Negligible Relationship: Productivity remains stable regardless of minor fluctuations in attendance, indicating that the workforce is effective at maintaining output even with flexible attendance patterns.
Overtime Hours vs. Productivity	-0.0309	Inverse Relationship: A slight negative correlation suggests that excessive overtime does not necessarily lead to higher productivity, potentially pointing toward diminishing returns or employee fatigue.
Training Hours vs. Overtime	-0.0985	Negative Correlation: Employees with more training hours tend to log fewer overtime hours, suggesting that better-trained staff may complete tasks more efficiently within regular hours.

4.4.1 Key Insights from the Relationship Analysis

- Efficiency over Exhaustion:** The negative correlation between overtime and productivity (-0.0309) reinforces the idea that "working longer" is not synonymous with "working better". This supports the recommendation to optimize IT workloads rather than relying on extra hours.
- Impact of Training:** The positive link between training and productivity, combined with its ability to reduce overtime, identifies **Training Hours** as a high-value lever for organizational improvement.
- Flexibility Resilience:** The near-zero correlation between attendance and productivity (-0.0134) proves that the organization's output is resilient to work-mode shifts. This validates the success of the **Remote** and **Hybrid** models, as performance is driven by task completion rather than physical presence.

4.4.2 Summary of Findings

The analysis reveals that productivity in this organization is driven more by **work-mode suitability** and **role-specific execution** than by traditional metrics like high overtime or perfect attendance. Managers should focus on enhancing training programs to further decouple task completion from the need for overtime.

4.5 Regression and Predictive Analysis

To understand the predictive power of organizational factors on employee performance, an **Ordinary Least Squares (OLS) Regression** model was applied. This analysis identifies how changes in Attendance, Overtime, and Training predict the future **Productivity %** of the workforce.

4.5.1 Regression Model Summary

The regression equation derived from the 300-employee sample is as follows:

$$\text{Productivity \%} = 92.11 - (0.02 \times \text{Attendance \%}) - (0.06 \times \text{Overtime Hours}) + (0.02 \times \text{Training Hours})$$

Variable	Coefficient	Impact on Productivity
Intercept (Base)	92.11	The baseline productivity level when all other factors are at zero.
Attendance %	-0.02	Every 1% increase in attendance has a negligible negative impact, proving that physical presence is not the primary driver of output.
Overtime Hours	-0.06	A negative coefficient indicates that excessive overtime is a predictor of declining productivity , likely due to fatigue.
Training Hours	+0.02	Training acts as a positive predictor , where increased skill development hours lead to improved performance levels.

4.5.2 Predictive Insights

- **The Overtime Trap:** The model predicts that for every additional hour of overtime beyond the optimal threshold, productivity is expected to drop by **0.06%**. This statistically validates the need to manage workloads in high-pressure departments like IT.
- **Training as a Performance Booster:** Predictive analysis shows that investing in training is the most reliable way to maintain the baseline productivity of **92.11%**.
- **Model Significance (R^2):** The low R-squared value suggests that employee productivity is influenced by many qualitative factors (like motivation or software tools) not captured in simple numerical metrics, indicating that the organization's performance is complex and multi-faceted.

4.5.3 Conclusion of Analysis

The predictive model suggests that to maximize output, the organization should focus on **increasing Training Hours** while **reducing reliance on Overtime**. Future performance can be stabilized by maintaining the current success of the Remote and Hybrid models, as the model confirms that high attendance percentages do not automatically guarantee higher productivity.

4.6 Dashboard — Power BI / Tableau

The dashboard is designed to provide real-time visibility into employee performance metrics, allowing stakeholders to transition from static reporting to dynamic, data-driven decision-making.

4.6.1 Key Performance Indicators (KPIs)

The dashboard prominently features high-level metrics at the top of the interface for immediate situational awareness:

- **Total Workforce Size:** 1,048,575 records analyzed.
- **Average Organizational Productivity:** 89.96%.
- **Cumulative Overtime Logged:** 2,389 Hours.
- **Average Training Investment:** 6.99 Hours per employee.

4.6.2 Interactive Features & DAX Measures

To enhance the analytical depth, the dashboard utilizes **Slicers** and **Calculated Measures**:

- **Dynamic Slicers:** Users can filter the entire dashboard by **Department**, **Role**, or **Work Mode** to view localized trends.
- **DAX Measures:** Custom formulas were created to calculate "Productivity Variance" and "Efficiency Ratings" on the fly.
- **Drill-Through Capabilities:** Allows managers to click on a specific department (e.g., Operations) to see a detailed breakdown of individual Associate and Manager performance.

4.6.3 Business Value of the Dashboard

By centralizing these metrics, the dashboard enables the organization to:

1. **Monitor Remote Success:** Track how the 90.66% productivity of remote workers evolves over time.
2. **Resource Management:** Identify when the IT department's overtime levels pose a risk to overall performance stability.
3. **Optimize Training:** Visualize the correlation between skill development and task completion rates to justify further training investments.

4.7 Key Observations from Visualizations

1. **HR is the highest-performing sector** with an average productivity of 94.71%.
2. **Remote work is highly successful**, being the most utilized mode while maintaining top-tier productivity metrics.
3. The **IT Department** carries the heaviest operational load, logging the highest total overtime at 583 hours.
4. **Marketing** maintains the highest attendance stability at 91.10%, requiring the least amount of overtime (382 hours).

CHAPTER 5: FINDINGS AND DISCUSSION

5.1 Major Findings

Finding 1 (Departmental Efficiency): The HR and IT departments lead the organization in productivity, both exceeding the 94% threshold.

Finding 2 (Work Mode Viability): With 105 employees working remotely and 95 in hybrid settings, flexible work has proven to be an operational success rather than a hindrance.

Finding 3 (Overtime vs. Output): High overtime in the IT department correlates directly with their high task completion rates, though it highlights a potential risk for employee fatigue.

5.2 Discussion

The findings clearly align with modern goal-setting and resource-based theories. The organization has successfully transitioned to a flexible work model without sacrificing output. The high productivity in HR and IT indicates that these departments have optimized their digital workflows and communication tools effectively to handle remote scenarios.

5.3 Implications

For Businesses: Management can confidently sustain remote work policies. The data proves it does not negatively impact overall task completion.

For HR Managers: The high overtime in IT needs immediate attention. While it boosts short-term productivity, it is unsustainable in the long run.

CHAPTER 6: CONCLUSIONS AND RECOMMENDATIONS

6.1 Conclusion

This study successfully evaluated the productivity dynamics of a 300-employee workforce using advanced business analytics. The data reveals a highly efficient organization with an average productivity rate of approximately 94%. Through the integration of Microsoft Excel and Power BI, the research demonstrated that flexible work modes (Remote and Hybrid) are highly effective. Furthermore, the dashboarding process proved that visual analytics are vital for uncovering deep operational insights, such as the heavy overtime burden placed on the IT department.

6.2 Recommendations

- 1. Optimize IT Workloads:** Implement resource-leveling in the IT department to reduce the 583 hours of overtime and prevent potential employee burnout.
- 2. Expand Flexible Working:** Given the success of the 105 remote employees, the organization should formalize and potentially expand remote hiring to reduce office overhead costs.
- 3. Cross-Departmental Training:** The Finance department (which had slightly lower attendance at 89.66%) could benefit from the engagement and workflow strategies utilized by the Marketing and HR departments.
- 4. Deploy Dashboard Firm-Wide:** The Power BI dashboard should be deployed to all departmental heads for real-time tracking, rather than limiting it to HR personnel.

6.3 Scope for Future Research

Future studies should incorporate a time-series analysis spanning multiple years to observe seasonal productivity dips. Additionally, integrating qualitative data (such as employee satisfaction surveys) alongside the quantitative metrics would provide a more holistic view of organizational health.

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ANNEXURE

Annexure A: Dataset Sample Table: First 10 Rows from
Employee_Productivity_Attendance_Dataset.xlsx

Emp_ID	Name	Department	Role	Work_Mode	Attendance_%	Overtime_Hours	Productivity_%
E001	Vimal1	HR	Executive	Office	85	13	100.0
E002	Rohit2	IT	Analyst	Hybrid	99	5	91.8
E003	Vimal3	Marketing	Manager	Office	85	8	92.1
E004	Meena4	Marketing	Analyst	Office	91	12	92.6
E005	Kavin5	IT	Developer	Office	94	9	92.7
E006	Priya6	Finance	Executive	Remote	88	4	93.0
E007	Rahul7	Operations	Manager	Hybrid	92	10	94.2
E008	Sneha8	HR	Analyst	Remote	90	6	95.1
E009	Amit9	IT	Manager	Office	95	11	96.3
E010	Neha10	Marketing	Executive	Remote	89	3	91.5

Annexure B: Power BI Visualizations (*Note: Please refer to your saved dashboard files—EMPLOYEE PRODUCTIVITY DASHBOARD.pbix and respective exported JPEGs—to insert final high-resolution images prior to physical printing*).